

What are the main determinants of social expenditure? A panel data approach for EU and OECD countries

Determinants
of social
expenditure

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Abstract

Purpose – The aim of this study is to analyze which socioeconomic factors (economic, demographic, and political) most commonly affect the social expenditure of the European Union (EU) and Organization for Economic Co-operation and Development (OECD) countries.

Design/methodology/approach – A panel data fixed-effects model is employed for 34 OECD and 23 EU countries between 2000 and 2020.

Findings – Results indicate that, in all country groups, economic factors have the most significant influence on social expenditures, with income being the primary determinant, particularly in EU countries. The negative impacts of unemployment and inflation underscore the importance of counter-cyclical measures adopted by countries to maintain stability in their social expenditures. The most influential demographic factor is found as the old-age-dependency ratio. While the rule of law affects social expenditure positively, government effectiveness and female labor force participation affect it negatively. The positive effect of Konjunkturforschungsstelle (KOF) indexes shows the globalization effect, which can be attributable to the compensation hypothesis.

Practical implications – Governments enforce inclusive and sustainable policies to boost economic activities and GDP, thus combating inflation and unemployment and regulating the labor market and socioeconomic problems about aging populations and women's economic participation to control social expenditures. The rule of law and institutional quality will also boost economic growth.

Originality/value – This study focuses on the effects of social expenditures in a broader view within the framework of the three main factors (economic, demographic, political) and attempts to determine the key factors that account for the differences in social expenditure between the OECD and EU countries.

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1. Introduction

Social expenditure (spending) is one of the main key components of fiscal policy that affects economic activities and promotes the welfare levels of citizens. It includes all public spending on social protection for employees, disadvantaged or vulnerable groups, education and health services. Social expenditure could be seen as a proxy for the sum of governmental outputs devoted to the tool of social amelioration (Castles, 2005). Social expenditures have been affected by countries' economic, demographic, political and institutional characteristics. The content and the extent of the social expenditures is one of the indicators of welfare and economic development level of a country. Due to the changing needs of the economy and society, countries are experiencing major pressures to expand the content and the extent of social expenditures throughout the years. Inevitable change has occurred in all countries



because of the 1970s' increasing reforms in the social policy in the developed countries, 1990s' combat with the challenges of an aging population and globalization and the 2000s' economic slowdowns, unemployment, political turmoil in the developing countries, and therefore increasing migration, poverty and the inequality and the effect of pandemic diseases since at the end of 2019. Although there are slight differences in countries, the importance of old-age spending, health and family or incapacity-related benefits has remained higher since 1980.

This study aims to analyze which socioeconomic (economic, demographic, and political) factors most commonly affect the social expenditures of 34 Organization for Economic Cooperation and Development (OECD) and 23 European Union (EU) countries [1]. A panel data fixed-effects model is employed for the period of 2000–2020. The main contribution of this study is to focus on the effects of social expenditures in a broader view within the framework of the impact of three major socioeconomic factors, and to compare any difference between OECD and EU countries including Türkiye as a candidate country. The paper is organized as follows: Sections 2 and 3 review the theoretical background and literature review. Section 4 displays the dataset, the methodology and discussion. Section 5 concludes the study.

2. Theoretical background: determinants of social expenditure

Determinants of social expenditure can be classified as economic, political and demographic factors.

2.1 Economic factors

The impacts of economic factors on social expenditure have been analyzed under globalization and its two opposite effects. The disciplining hypothesis states that globalization restrains social expenditures and decreases the size and extent of government welfare programs due to the restrictive effects of international competition. The efficiency hypothesis states the opposite relationship between social expenditure and openness. Globalization imposes the market interest in governments rather than social issues. Hence, social expenditures reduce. The compensation hypothesis suggests the positive effects of globalization on social expenditures. Economic actors that are negatively affected by globalization can be compensated through an increase in social welfare programs (Dreher *et al.*, 2008). Through the impact of globalization, governments can increase social expenditures to increase workers' productivity and overcome the risks and volatility of economics (Kaufman and Segura-Ubiergo, 2001; Martín-Mayoral and Sastre, 2017) because the public investments that support human capital will increase the productivity and the quality of the workforce that which is necessary for the international competition and the business sector. Kaufman and Segura-Ubiergo, 2001 state that the increase in the international competitiveness of countries may cause uncertainty, unequal distribution of socioeconomic factors and social dislocation. The macroeconomic factors used as a proxy for globalization are openness to trade and capital openness. These are measured by the sum of outflows and inflows of foreign direct investment (FDI) (as a share of GDP) or FDI stocks, the restrictions on capital account transactions and the Konjunkturforschungsstelle (KOF) Index of Globalization developed by Dreher (2006a) (Kaufman and Segura-Ubiergo, 2001; Garrett and Mitchell, 2001; Dreher *et al.*, 2008; Haelg, 2020). Globalization and its three components are economic, social and political globalization. They are measured by using the Swiss Economic Institute (Konjunkturforschungsstelle—KOF) indices of globalization Gygli *et al.* (2019). The index range from 1 (least globalized) to 100 (most globalized) for each dimension. Economic globalization is characterized as the flows of goods, services, capital and information abroad, and perceptions concomitant with market exchanges. The KOF index of economic globalization comprises information on economic flows such as trade, portfolio

investment, FDI and restrictions on international trade, capital movements and investment (Santos and Simões, 2021). While social globalization is expressed as the mobility of people and the spread of information, ideas, images; political globalization is characterized by a diffusion of government policies (Dreher, 2006a).

The KOF index of political globalization is calculated based on the number of international organizations to which the country belongs, the number of international treaties that have been signed and the number of embassies in a country, as well as its participation in the missions of the UN Security Council. The index captures the extent of political cooperation and integration among countries. The KOF social globalization index comprises information on personal contacts and considers connections between foreign country citizens (Santos and Simões, 2021). It measures the degree of cultural, informational, and personal exchange between countries. Some researchers have evaluated the effects of global indices (economic, political and social) on social expenditures in terms of welfare regimes (Kim and Zurlo, 2009; Jeong, 2010; Leibrecht *et al.*, 2011; Onaran and Boesch, 2014; Yay and Aksoy, 2018; Santos and Simões, 2021). As stated in the study of Görg *et al.* (2009), as a public good, improving human capital through social expenditure in education and health standards fulfills affirmative contribution to attracting FDI. Human capital investments boost the skills and abilities of the workforce, thus increasing productivity, technological progress, economic growth, welfare of countries (Karaalp-Orhan, 2018). To sum up, compensation hypothesis indicates a positive relationship between social expenditure and globalization.

Economic growth (GDP per capita growth) and the welfare level of countries affect the extent of social expenditures. Wagner's Law (Wagner, 1883) states that during the development of the countries, the share of public spending in national income tends to increase. Because public intervention might manage natural monopolies due to the expansion of expenditures on culture and welfare after development and public functions substitute private activities (Magazzino *et al.*, 2015). The size of GDP and countries' economic performance affect social expenditure (Castles, 2005). A bidirectional causal relationship can be found between economic growth (business cycles) and social expenditure. While social expenditures contribute to the countries' economic growth through improvements in health, education (Aban and Garcia-Vigonte, 2022), reduction of poverty and inequality and attracting FDI, positive economic growth will increase social expenditures and the welfare level of countries. Conversely, the unequal distribution of income affects social expenditures negatively (Molina-Morales *et al.*, 2011). Governments use social expenditure programs to eliminate income inequality among different income groups through pensions, unemployment insurance, housing, social protection, health and education (Verberi and Yaşar, 2021). Growth in total income may reduce the government's budget constraint and enables the government to raise public expenditure (Schuknecht, 1998). A positive relationship is observed between tax revenue and GDP per capita. Through the increase of GDP and economic development level, governments increase their social expenditure to satisfy the needs of society (Magazzino, 2016; Martín-Mayoral and Sastre, 2017). Besides, social expenditures increase with the increase in per-capita income, expressed by Wagner's Law (Garrett and Mitchell, 2001; Kaufman and Segura-Ubiergo, 2001; Avelino *et al.*, 2005). However, the Keynesian approach states that social expenditures increase efficiency and productivity and affect growth positively. A controversial effect of economic growth on social expenditure can be observed by Avelino *et al.* (2005), Leibrecht *et al.* (2011), Schuknecht and Zemanek (2018), Martin and Algarhi (2023). Aban and Garcia-Vigonte (2022) state that contrary to short-term, long-term social spending on education and health helps economic growth significantly. Dreher (2006b) states that social expenditures decrease in periods of low economic growth. Namely, social expenditures decrease when the economy goes sour (Castles, 2005). Governments cut all expenditure programs in recession times to balance the budget expenditures (Snyder and Yackovlev, 2000; Kittel and Obinger, 2003). A positive

correlation can be found between recessions and social expenditures. While expenditures can be expected to grow in periods of recession, they may be likely to decline in times of prosperity. It can be stated that the correlation between economic performance and social expenditure is sensitive and has an anticyclical character (Castles, 2005).

During recessions (high unemployment rate periods), governments are likely to increase social expenditure to decrease the adverse effects of the economic crisis, boost economic activities and promote employment generation (Martín-Mayoral and Sastre, 2017). The high unemployment rate is related to slow economic growth, resulting in the decline of occupational structures. Countries introduce social expenditure programs such as unemployment insurance. Studies generally find a positive relationship between the unemployment rate and social expenditures (Snyder and Yackovlev, 2000; Kittel and Obinger, 2003).

Inflation is another economic factor that affects social expenditure. Aubin *et al.* (1988) state that public capital expenditure is likely to decrease when inflation increases. Lin (1992) mentions the negative effect of inflation on the growth rate of government expenditure. Inflation causes a reduction in tax revenues, so to eliminate inflation, government expenditures are reduced by government thereby, government-fixed nominal expenditure levels are determined according to the estimations of desirable debt level and the tax revenues at the beginning of the fiscal year. Soroka *et al.* (2006) state the positive correlation between inflation and social expenditures. Martín-Mayoral and Sastre (2017) mention the public indebtedness effect on social expenditures. Lora and Olivera (2007) state that social expenditures decrease as a result of high-interest payments on debt and a high debt ratio.

The female labor force participation rate (FLFPR) is a socioeconomic and demographic factor for social expenditures and has explicit and implicit effects on the labor market. The FLFPR effect on social expenditures has controversial results. Changes in labor force participation have significant economic and budgetary implications. The direct effects occur according to social welfare state policy in general and social service. Those policies might change women's and men's attitudes. Female labor supply is not only a driver and an outcome of development and growth but also boosts the reduction of child poverty and the overall poverty of the countries (Karaalp-Orhan, 2017). Women's economic contribution and the enhancement of family income will promote preventing poverty (also women/children) and allocating/reducing social expenditures significantly. The indirect effect occurs when women participate in unions, parties or organizations (Woittiez and Kapteyn, 1998). FLFP will cause to increase women's demand for a more comprehensive welfare state and will increase social expenditures for the medium and long-term period. Allik (2016), Ennsner-Jedenastik (2017), Park (2017) and Detraz and Peksen (2018) found a positive correlation between women's political representation and social expenditures (welfare spending) and FLFP. The higher FLFP may increase labor market-related expenditures such as employment insurance, maternity leave, training, severance pay, and leave for a short period.

2.2 Demographic factors

Demographic structure (and determinants) is the demand-side factor that directly affects the extent of social expenditure. Hereunder, changes in the dependency ratio will provide information about the potential needs and requirements of the transformation in the population age structure. Age-related expenditures (pensions, social care and geriatric health expenditures) are the main components of social expenditure and are positively related to age. Hence, the aging population tends to lead to an increase in social expenditure. Economists focused on the impacts of changing population age groups on social expenditures. Although many developed countries experience a rise in the old-age population relative to the tax-payer age groups and a decline of the proportion of young people, developing countries still

experience a high level of young population. Thereby, the effects of the age group on social expenditure can be different. For instance, an older population might tend to demand more government contribution to pensions, health and welfare programs for elderly care (Lindert, 1996; Castles, 2005; Asher and Zen, 2016). Lindert (1994, 1996) state a strong positive relationship between the rise of the size of the old-age population and social expenditures, especially on pensions. Baldacci and Lugaes (1997) state that the aging population will increase the fiscal burden in the next decades. A higher life expectancy in old age causes a shift from the mean age to the elderly age of the population at which the health services are heavily demanded. Contrary, a younger population might tend to demand more government contribution to childcare, daycare center, youth centers, libraries, schools, kindergartens and pediatric hospitals.

2.3 Political factors

A vast study analyzes the effects of the governments' political ideology and their impacts on social expenditure [2]. Political cycle theories analyze the government incentives effects on policy choices. Opportunistic (electoral) theories state that, regardless of ideological differences, governments will try to influence economic policies during elections to increase their chances of being reelected (Bove *et al.*, 2017). Due to policymakers having a direct and immediate impact on voters' welfare, they prefer to use the rise of public expenditure as a tool before the elections (Schuknecht, 1998). Vergne (2009) states that rather than capital expenditures, public expenditures mainly concentrate on social expenditures such as wages and subsidies during the election years. According to partisan theories, right-wing governments focus on less income redistribution and less expansionary policies than left-wing governments. However, no consensus is found for governments' most preferred policy tools to accomplish political aims. Regime types also affected social expenditures. Martín-Mayoral and Sastre (2017) state that democratic regimes have higher social costs than authoritarian regimes due to high electoral risks.

The efficient allocation of government resources according to the needs of citizens and how well the government policies achieve desired objectives also affected social expenditures. World Bank (2023a) states that government effectiveness involves the quality of policy implementation and formulation, public and civil services quality, the degree of independence from political pressures and the credibility of the government's commitment to such policies. Government effectiveness represents clean water and electricity all over the country, railways and airports for transportation, adequate and equal access to education and health system, poverty reduction, quality of bureaucracy and stability in political processes. When government efficiency is associated with the level of development and welfare in a country (Knack and Keefer, 1995; La Porta *et al.*, 1999), it can be expected that there will be a negative relationship with social expenditures, just like income growth. The rule of law directly affects the economic structure, governmental regulations and social life of the citizens. The rule of law is one of the prominent components to measure economic freedom, and it also affects economic growth positively. The presence of the rule of law is essential for the efficient distribution of resources, and it boosts the functioning of the economy and forms the basis of socioeconomic development and welfare. World Bank (2023b) states the rule of law as the perceptions of the extent to which representatives trust and abide by the rules of society and especially the quality of the implementation of the property rights, contracts, the courts, the police and the likelihood of violence and crime. The rule of law will bring democracy to a country and positively affect social expenditures similarly. Huber *et al.* (2008) and Martín-Mayoral and Nabernegg (2014) state the positive effect of democratic regimes on social expenditures; Tomita and Kimura (2021) state that countries, where the rule of law is recognized, tend to increase their social security expenditures. IMF (2020) states that the rule of

law and government effectiveness are the prominent factors in the efficiency of social expenditure. Government effectiveness, corruption control and the rule of law are important indicators of institutional quality. A positive relationship is stated between government expenditure efficiency and institutional quality. Stronger accountability and transparency over public resource use promote efficiency and minimize wasteful spending. Political instability, violence and terrorism negatively affect the distribution of government resources and create uncertainties. Uncertainties cause weakness in legal regulations and decrease investments due to a drop in the expected return on investments (Asteriou and Price, 2001, p. 386) and also affected government policies and increased public expenditures. Political stability affects economic growth positively because it provides confidence for investors and prosperity for the citizens (Shabbir *et al.*, 2016), and social expenditures will decrease through the indirect growth channel.

3. Literature review

Various empirical studies have analyzed the determinants of social expenditures by using different macroeconomic variables. Kaufman and Segura-Ubiergo, 2001 find the negative effect of foreign trade and the positive impacts of GDP per capita, the old-age population and democratic processes on social expenditure. Garrett and Mitchell (2001) identify a positive relationship between the unemployment rate and the dependent population. Castles (2005) finds the impact of economic growth and no connection between party ideology and expenditure. Avelino *et al.* (2005) conclude the negative and positive effects of GDP per capita, inflation; urban population, dependency ratio, unemployment and democracy on social expenditure. Dreher *et al.* (2008) conclude that government expenditures do not influence by globalization [3]. Jeong (2010) analyzes whether economic globalization causes negative pressure on social policies. Leibrecht *et al.* (2011) state that the KOF overall globalization index shows a positive relationship with social expenditure in conservative welfare regimes; a negative relationship is found in Eastern European and social-democratic countries. Economic growth, inflation and country size affect social expenditures negatively; dependent population affects positively. Molina-Morales *et al.* (2011) find a positive relationship between the elderly population, income and social expenditures. Martín-Mayoral and Nabernegg (2014) indicate a positive and higher impact of unemployment and democracy. Martín-Mayoral and Sastre (2017) find the fiscal burden effect, GDP per capita, FDI inflow and international trade on social expenditure and justify the compensation hypothesis. Bove *et al.* (2017) state that military expenditure is higher in the rule of right-wing administrations, and social expenditures are higher under the rule of left-wing administrations. Schuknecht and Zemanek (2018) identify the negative effects of the income and government debts, the positive effects of the old-age dependency ratio, the unemployment rate and dominant political ideologies. Kaymaz (2018) found the negative, positive and prominent effects of the inflation rate, unemployment and dependent population. Santos and Simões (2021) focus on globalization and social expenditure for OECD countries by considering the welfare state models. Aban and Garcia-Vigonte (2022) find the positive impacts of long-term social spending on education, health on economic growth. Martin and Algarhi (2023) state no significant effect of social spending on economic growth. Çelikay (2023) finds the progressive effect of social expenditures on chronic unemployment in generous-spending countries. Previous studies use different macroeconomic data and county groups for various years and periods to analyze the determinants of social expenditures. Therefore, a consensus is not found for the signs of the determinants.

4. Empirical model and data

Potential determinants of social expenditure have cross-country effects. Panel data can help us find the relationship between variables by giving more informative data, more variability,

less collinearity among the variables, more degrees of freedom and more efficiency rather than time-series data (Baltagi, 2008). In the study, a panel data model is employed to find the relationship between social expenditure and its determinants according to the following regression.

$$SOCX_{i,t} = \alpha_{i,t} + \beta_1 ECON_{i,t} + \beta_2 DEM_{i,t} + \beta_3 POL_{i,t} + u_i + \varepsilon_{i,t} \quad (1)$$

where the SOCX is a total social expenditure, ECON is the economic determinants, DEM represents demographic determinants, and POL refers to political variables. u_i is the unobserved country-specific effects, $\varepsilon_{i,t}$ is the error term. The subscripts i and t represent the country and time period, respectively.

4.1 Data and limitations

Economic determinants (ECON) comprise GDP per capita (GDP_pc), consumer price index (CPI), unemployment rate (UNR) and female labor force participation rate (FEM), KOF Globalization Index, KOF Economic Globalization Index (KOF_eco), KOF Social Globalization Index (KOF_soc) and KOF Political Globalization Index (KOF_pol). Demographic determinants (DEM) include the young-age population (AGE_young) and the old-age population (AGE_old). Political variables (POL) represent the rule of law, political stability and absence of violence/terrorism and government effectiveness index (see Table 1).

The dataset comprises unbalanced annual macroeconomic data for 34 OECD and 23 EU countries. Due to data constraints, Colombia, Costa Rica, Japan and Australia are excluded

Variable type	Code	Description	Source
Dependent variable	SOCX	Social Expenditure (% of GDP)	OECD Social Expenditure Database https://www.oecd.org/social/expenditure.htm
Independent variables	GDP_pc	GDP per-capita (constant 2015)	World Bank
	CPI	Consumer Price Index	WDI https://databank.worldbank.org/source/world-development-indicators
	UNR	Unemployment, total (% of the total labor force) (modeled ILO estimate)	
	FEM	The labor force participation rate, female (% of female population ages 15+) (modeled ILO estimate)	
	AGE_old	The ratio of older dependents—people older than 64—to the working-age population—those ages 15–64	
	AGE_young	The ratio of younger dependents—people younger than 15—to the working-age population—those ages 15–64	
	GOV_Eff	Government Effectiveness Index	
	POL.Stab	Political Stability and Absence of Violence/Terrorism	
	RULE_Law	Rule of law	
	KOF index	Globalization index	KOF Swiss Economic Institute https://kof.ethz.ch/en/forecasts-and-indicators/indicators/kof-globalisation-index.html
	KOF_eco	Economic Globalization index	
	KOF_soc	Social Globalization index	
	KOF_pol	Political Globalization index	

Source(s): Authors' compilations

Table 1.
Definitions and the
sources of the variables

from the OECD dataset; Bulgaria, Cyprus, Malta, Romania and Croatia are excluded from the EU dataset. The United Kingdom is included despite Brexit. The main descriptive statistics for the dependent and explanatory variables are defined in [Table 2](#).

The Stata 15.0 program is used for the analysis. Diagnostic tests for model estimation are shown in [Table 3](#).

The robustness tests included heterogeneity, cross-section dependence and second-generation unit root tests, respectively. [Pesaran and Yamagata \(2008\)](#) test results show that the slope coefficients were heterogeneous. The cross-sectional dependence [Pesaran \(2004\)](#) test found cross-sectional dependence for all countries groups. The stationarity (unit-root) test results show that variables contain an I(0) unit root and become stationary at the first difference I(1), except AGE_young and AGE_old variables for the EU23 + TR. The Hausman test, which is based on the assumption that the fixed effects model is consistent and unbiased, is applied for model selection, and the fixed effects estimator is found for the models. According to the Wald test (heteroscedasticity test) results, the null hypothesis is rejected, and error term variances are not constant for all observations. The [Baltagi and Wu \(1999\)](#) autocorrelation test results show the null hypothesis is rejected (for all models, LBI <2), and autocorrelation is found between error terms. Therefore, the estimations include robust estimation results where the heteroscedasticity effect and autocorrelation problems are eliminated ([Hayes and Cai, 2007](#)). The robust standard errors of the estimation results are also shown in parentheses in [Table 4](#).

4.2 Discussion

The findings in [Table 4](#) indicate that the factors that affect the social expenditure in OECD and EU countries are mainly economic factors. Income is the most prominent factor affecting social expenditure in all groups. However, it is more effective on the social spending of EU countries than OECD. This can be explained by the relatively more generous EU social policies. A slight reduction of the income effect is observed in Model III and IV, where Türkiye is included. A negative and statistically significant relationship is found between income and social expenditures in all models (counter-cycle). That is in line with the results of [Castles \(2005\)](#), [Soroka et al. \(2006\)](#), [Martín-Mayoral and Sastre \(2017\)](#) and [Haelg \(2022\)](#). While the size of GDP per capita and countries' economic performance affect social expenditure, social expenditure can be higher in the recession period than in times of prosperity ([Castles, 2005](#)). This indicates that countries implemented anticyclical precautions to stabilize their social

Variables	Mean	Standard deviation	Min	Max	Obs
SOCX	19.56	5.94	4.39	32.03	680
GDP_pc	34386.39	21523.73	6076.00	105454.70	680
CPI	97.75	16.34	20.59	234.44	680
UNR	7.71	4.16	1.81	27.47	680
FEM	52.55	8.15	22.88	73.82	680
AGE_old	15.28	3.74	5.15	23.01	680
AGE_young	17.96	4.06	12.45	34.20	680
GOV_Eff	85.22	11.35	39.90	100.00	680
POL_Stab	71.95	22.30	4.76	100.00	680
RULE_Law	84.24	14.69	27.40	100.00	680
KOF_Index	80.52	6.81	58.00	91.00	680
KOF_eco	75.19	9.70	46.00	93.00	680
KOF_soc	79.98	8.02	47.00	92.00	680
KOF_pol	86.43	10.38	51.00	98.00	680

Table 2.
Descriptive statistics

Source(s): Authors' calculations

								Determinants of social expenditure
		EU23		EU23 + TR		OECD34		
		Test- statistics	p-value	Test- statistics	p-value	Test- statistics	p-value	
Homogeneity	Delta	3.527*	0.000	1.999**	0.046	2.526**	0.012	
Pesaran and Yamagata (2008)	Adj. Delta	5.577*	0.000	3.293*	0.001	4.162*	0.000	
	Delta	2.659	0.008	1.390	0.164	1.829	0.067	
Cross-sectional dependence	Adj. Delta	4.854	0.000	2.710*	0.007	3.565	0.000	
	CD-	16.013*	0.000	12.752*	0.000	12.405*	0.000	
	Pesaran (2004)	15.371*	0.000			8.987*	0.000	
Second generation unit root test	t-bar	p-value		t-bar	p-value	t-bar	p-value	
I(0)	SOCX	−1.200	0.996	−1.405	0.946	−1.386	0.975	
	LnGDP_	−1.614	0.747	−1.457	0.913	−1.452	0.944	
	pc							
	CPI	−1.393	0.960	−1.172	0.997	−1.050	1.000	
	UNR	−1.504	0.885	−1.395	0.951	−1.533	0.870	
	AGE_	−0.441	1.000	0.088	1.000	−0.131	1.000	
	young							
	AGE_old	−2.636*	0.000	−2.733*	0.000	−2.008***	0.056	
	FEM	−1.886	0.253	−1.831	0.330	−1.878	0.199	
	GOV_Eff	−2.044***	0.075	−2.184**	0.017	−2.255*	0.001	
	POL_	−2.319*	0.003	−2.192**	0.015	−2.233*	0.002	
	Stab							
	RULE_	−1.879	0.264	−2.031***	0.081	−1.890	0.180	
	Law							
	KOF_	−2.535*	0.000	−2.355*	0.002	−2.409*	0.000	
	Index							
	KOF_eco	−2.329*	0.002	−2.167**	0.020	−2.183*	0.005	
	KOF_soc	−2.756*	0.000	−2.895*	0.000	−2.917*	0.000	
	KOF_pol	−2.637*	0.000	−2.616*	0.000	−2.417*	0.000	
	SOCX	−3.286*	0.000	−3.073*	0.000	−2.920*	0.000	
I(1)	LnGDP_	−3.297*	0.000	−2.828*	0.000	−2.844*	0.000	
	pc							
	CPI	−2.476*	0.000	−2.114**	0.036	−2.488*	0.000	
	UNR	−2.950*	0.000	−2.827*	0.000	−2.951*	0.000	
	AGE_	−2.570*	0.000	−1.457	0.913	−1.185*	0.000	
	young							
	AGE_old	−2.437*	0.000	−0.973	1.000	−0.690*	0.000	
	FEM	−4.248*	0.000	−3.936*	0.000	−3.939*	0.000	
	GOV_Eff	−4.846*	0.000	−4.709*	0.000	−4.833*	0.000	
	POL_	−4.755*	0.000	−4.510*	0.000	−4.570*	0.000	
	Stab							
	RULE_	−4.480*	0.000	−4.198*	0.000	−4.323*	0.000	
	Law							
	KOF_	−4.831*	0.000	−4.779*	0.000	−4.758*	0.000	
	Index							
	KOF_eco	−4.176*	0.000	−4.055*	0.000	−4.238*	0.000	
	KOF_soc	−4.682*	0.000	−4.594*	0.000	−4.857*	0.000	
	KOF_pol	−4.767*	0.000	−4.935*	0.000	−4.439*	0.000	
Hausman test	Model 1	67.20*	0.000	65.78*	0.000	68.20*	0.000	
	Model 2	118.49*	0.000	116.23*	0.000	89.75*	0.000	
Wald test	Model 1	608.85*	0.000	861.34	0.000	2488.41*	0.000	
	Model 2	793.77*	0.000	1189.57	0.000	3555.25*	0.000	
Baltagi and Wu (1999) LBI test	Model 1	0.744		0.621		0.612		
	Model 2	0.766		0.620		0.621		
Note(s): *, **, ***, %1, %5, %10								Table 3. Diagnostic tests
Source(s): Authors' calculations								

Models	EU23 I	EU23 II	EU23 + TR III	EU23 + TR IV	OECD34 V	OECD34 VI
LnGDP_pc	-9.79* [2.24]	-9.56* [2.19]	-8.37* [2.36]	-8.24* [2.68]	-6.52* [2.36]	-6.57** [2.57]
UNR	0.14** [0.05]	0.11** [0.05]	0.18* [0.04]	0.14* [0.04]	0.23* [0.04]	0.20* [0.04]
CPI	0.13* [0.04]	0.12* [0.04]	0.06* [0.02]	0.05* [0.02]	0.04* [0.01]	0.04* [0.01]
FEM	-0.15*** [0.08]	-0.22** [0.08]	-0.09 [0.07]	-0.18** [0.08]	-0.05 [0.06]	-0.13** [0.06]
AGE_old	0.22 [0.20]	0.18 [0.19]	0.29*** [0.16]	0.23 [0.18]	0.42* [0.14]	0.38* [0.14]
AGE_young	0.13 [0.21]	0.23 [0.21]	0.06 [0.22]	0.15 [0.20]	-0.10 [0.15]	-0.09 [0.15]
GOV_Eff	-0.05 [0.04]	-0.05 [0.04]	-0.08** [0.03]	-0.07** [0.03]	-0.04 [0.03]	-0.05*** [0.03]
POL_Stab	-0.03*** [0.02]	0.03*** [0.01]	-0.04* [0.01]	-0.03** [0.01]	-0.03* [0.01]	-0.03* [0.01]
RULE_Law	0.14** [0.07]	0.12*** [0.06]	0.15* [0.05]	0.12** [0.06]	0.15* [0.05]	0.13* [0.05]
KOF_Index	0.06 [0.10]		0.20* [0.07]		0.11** [0.06]	
KOF_eco		-0.08 [0.05]		-0.07 [0.05]		-0.07** [0.03]
KOF_soc		0.16** [0.07]		0.21* [0.04]		0.16* [0.05]
KOF_pol		0.01 [0.05]		0.08*** [0.04]		0.04 [0.03]
cons	99.55* [19.77]	99.95* [20.03]	78.47* [21.65]	82.04* [24.41]	62.92* [20.09]	68.73* [21.71]
N	483	483	480	480	680	680
R-sq	0.57	0.59	0.56	0.61	0.55	0.58
Time	2000–2021		2000–2019		2000–2019	

Table 4.
Results of panel data
with fixed effects
robust estimation

Note(s): Robust standard errors [] in parentheses, * $p < 0.01$, ** $p < 0.05$, *** $p < 0.10$

Source(s): Authors' calculations

expenditures. The economic crises and their socioeconomic consequences contributed to this process.

Other macroeconomic variables that affect social expenditures in all models are CPI and the unemployment rate. As found in the study of [Soroka *et al.* \(2006\)](#) and [Magazzino, 2016](#), a positive and statistically significant relationship is found between social expenditures and inflation. [Opler's \(1988\)](#) inflation increases public outlay (IPO) hypothesis emphasizes that the efforts of groups who suffer losses from inflation to recoup their losses may cause government expenditure to increase with inflation. Inflationary expectations can boost the government to raise its expenditures in general terms. Inflation is more effective on the social expenditure of EU countries than OECD countries.

The impact of the unemployment rate is found positive and statistically significant in all models. That is in line with the results of [Snyder and Yackovlev \(2000\)](#), [Kittel and Obinger \(2003\)](#), [Avelino *et al.* \(2005\)](#), [Soroka *et al.* \(2006\)](#), [Martín-Mayoral and Nabernegg \(2014\)](#), [Magazzino \(2016\)](#) and [Çelikay \(2023\)](#). When the unemployment rate increases, it is commonly expected that unemployment-related expenditures should automatically increase. This shows the government's efforts to compensate for and even eliminate the destructive effects of crisis, boost economic activities and to promote employment generation.

Unlike the previous studies, a negative relationship was found between FLFPR and social expenditures for all models. Women and children are considered to be in the disadvantaged group, so they do not get a sufficient share of the income distribution in the economy and are poorer than men. Therefore, women's employment will increase family income and help to combat the reduction of women's and child poverty. FLFPR is more effective on the social expenditure of EU countries than OECD countries. The relatively more generous social policies for women will influence the allocation of social expenditures.

Age-related expenditures take a prominent place for social expenditures. The main components of social expenditure, such as health expenditures, pensions and social care services, are strongly positively related to age. A positive relationship is found between AGE_old and social expenditure in OECD countries; however, no significant relationship is found in EU countries. The indicator for old age is found statistically significant for Model III, V and VI. Old-age populations are more effective in the social expenditure of OECD countries than EU countries. Also stated in [Lindert \(1994, 1996\)](#), [Baldacci and Lugares \(1997\)](#), [Soroka et al. \(2006\)](#), [Leibrecht et al. \(2011\)](#), [Molina-Morales et al. \(2011\)](#), [Schuknecht and Zemanek \(2018\)](#), [Haelg \(2022\)](#) aging population increases social expenditures. Additionally, we did not identify a significant relationship between the (AGE_young) and social expenditure in any model. The impact of the old-age-dependency ratio on social expenditure is found more than the young-age-dependency ratio.

Government effectiveness has a negative effect on social expenditures when considering the increase in government effectiveness in terms of the development and welfare level of the country ([Knack and Keefer, 1995](#); [Kaufmann et al., 1999](#); [La Porta et al., 1999](#)), a result similar to the income increase has been obtained. Like social expenditures reduction when the GDP rise, social expenditures decrease when government effectiveness increase. An efficient government can allocate resources according to the requirements and wants of the economy, thus boosting economic growth. Government effectiveness is found more effective on the social expenditure of OECD countries than EU countries. Political stability has a negative and statistically significant impact on social expenditures in all models. Political stability affects economic growth positively because it provides confidence for investors and prosperity for the people in a country ([Shabbir et al., 2016](#)), thus social expenditures may decrease through the indirect growth channel. Rule of law has a positive and significant effect on social expenditures where it is found more effective on the social expenditure of OECD countries than EU countries. When we associate the rule of law with democracy, results compatible with the literature were obtained ([Jeong, 2010](#); [Martín-Mayoral and Nabernegg, 2014](#)).

In this study, the impact of globalization has been discussed in two ways. First, the KOF index of globalization effect on social expenditures. It is found positive and statistically significant for Model I and V ([Kaufman and Segura-Ubiergo, 2001](#); [Avelino et al., 2005](#); [Leibrecht et al., 2011](#); [Santos and Simões, 2021](#)). Second, are the economic, social and political global index effects. The effect of the economic global index on social expenditures is negative and statistically significant for Model VI. Similarly, [Kim and Zurlo \(2009\)](#) also find a negative relationship between social expenditure and economic globalization for the liberal, social democratic and conservative regimes. The social global index effect on social expenditures is positive and statistically significant for Model II, IV and VI. The impact of the political global index on social spending is positive and statistically significant only for Model IV.

5. Conclusion

This study aims to investigate which economic, demographic, and political factors mainly explained differences in social expenditures of OECD, EU countries and Türkiye as a candidate country of the EU.

According to the results, economic factors are the most influential indicator that affects the social expenditures of the EU and OECD countries. While income is the most prominent factor affecting social expenditures in all country groups, it is more effective on the social spending of EU countries than OECD. This can be explained by the relatively more generous social policies in the EU. The unemployment rate affected social expenditures positively. This shows the rise of unemployment-related expenditures. As the third factor, a positive effect of inflation on social expenditure was found. Inflation can boost the government to increase its expenditures. Generally, this situation can be attributed to the “inflation increases public outlay” hypothesis for all country groups. The impact of economic factors on social expenditures can be considered as the governments’ implementations to counteract the negative effects of the economic crises in the last decades, such as the 2008 international financial crisis and its socioeconomic consequences. Contrary to the previous studies, a negative relationship was found between FLFPR and social expenditures for all models. Women’s participation in the economy will increase family income and contribute to the decrease of women’s and child poverty, thus, affecting social expenditure negatively. The most influential demographic factor is found as the old-age dependency ratio. The aging population increases social expenditures. Among the political variables, the rule of law has a positive effect on social expenditure, whereas government effectiveness and political stability have negative effects on social expenditure. Institutional quality will boost economic growth through the efficient allocation of resources, therefore provide to decrease the dependency of vulnerable groups on social expenditures. The KOF index shows the positive impact of globalization on the social expenditure of EU and OECD countries. This indicates the evidence in favor of the compensation hypothesis.

Social expenditure is commonly accepted as a necessary policy instrument to support and promote inclusive long-term economic growth, reducing poverty, inequality and welfare. Governments enforce inclusive and sustainable policies to boost economic activities and GDP, thus combatting inflation and unemployment and regulating the labor market and socioeconomic problems about aging populations to control social expenditures. The labor market regulations of women’s participation in the economy, the improvements and transparent administrations for the rule of law, and institutional quality will boost economic growth.

Moreover, the country-specific structural differences, global economic crises, international migration that has stemmed from civil and regional wars, and pandemic diseases such as the COVID-19 that happened in the last years increase the number of disadvantaged groups in need of protection, and these conditions directly affect social expenditures. Due to the data limitations, this study does not comprise all these factors but can be included in further studies.

Notes

1. Due to data constraints, while Colombia, Costa Rica, Japan and Australia exclude from the OECD dataset, and Bulgaria, Cyprus, Malta, Romania and Croatia exclude from the EU dataset, the United Kingdom is included despite Brexit.
2. See the summary of earlier literature in the study of [Martín-Mayoral and Sastre \(2017\)](#).
3. See the summary of earlier literature about the impact of globalization on government expenditure in the study of [Dreher et al. \(2008\)](#).

References

- Aban, M.P. and Garcia-Vigonte, F. (2022), “A study on the effects of social spending on economic activity”, SSRN, available at: <https://ssrn.com/abstract=4123354> or <http://dx.doi.org/10.2139/ssrn.4123354> (accessed 30 May 2022).

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- Allik, M. (2016), "Welfare state and representation: do women make the welfare state or does the welfare state make women representatives?", *Representation*, Vol. 52 No. 4, pp. 253-269, doi: [10.1080/00344893.2017.1300599](https://doi.org/10.1080/00344893.2017.1300599).
- Asher, M.G. and Zen, F. (2016), "Age-related pension expenditure and fiscal space: an overview", in Asher, M.G. and Zen, F. (Eds), *Age-Related Pension Expenditure and Fiscal Space Modelling Techniques and Case Studies from East Asia*, Routledge, New York, pp. 1-13.
- Asteriou, D. and Price, S. (2001), "Political instability and economic growth: UK time series evidence", *Scottish Journal of Political Economy*, Vol. 48 No. 4, pp. 383-399, doi: [10.1111/1467-9485.00205](https://doi.org/10.1111/1467-9485.00205).
- Aubin, C., Berdot, J.P., Goyeau, D. and Lafay, J.D. (1988), "The growth of public expenditure in France", in Lybeck, J.A. and Henrekson, M. (Eds), *Explaining the Growth of Government*, Amsterdam North-Holland, pp. 201-230.
- Avelino, G., Brown, D.S. and Hunter, W. (2005), "The effects of capital mobility, trade openness, and democracy on social spending in Latin America, 1980-1999", *American Journal of Political Science*, Vol. 49 No. 3, pp. 625-641, doi: [10.1111/j.1540-5907.2005.00146.x](https://doi.org/10.1111/j.1540-5907.2005.00146.x).
- Baldacci, E. and Lugares, S. (1997), "Social expenditure and demographic evolution: a dynamic approach", *Genus*, Vol. 53 Nos 1/2, pp. 61-78.
- Baltagi, B.H. (2008), *Econometric Analysis of Panel Data*, John Wiley & Sons, Chichester.
- Baltagi, B.H. and Wu, P.X. (1999), "Unequally spaced panel data regressions with Ar(1) Disturbances", *Econometric Theory*, Vol. 15 No. 6, pp. 814-823, doi: [10.1017/s0266466699156020](https://doi.org/10.1017/s0266466699156020).
- Bove, V., Efthymioulou, G. and Navas, A. (2017), "Political cycles in public expenditure: butter vs guns", *Journal of Comparative Economics*, Vol. 45 No. 3, pp. 582-604, doi: [10.1016/j.jce.2016.03.004](https://doi.org/10.1016/j.jce.2016.03.004).
- Castles, F.G. (2005), "Social expenditure in the 1990s: data and determinants", *Policy and Politics*, Vol. 33 No. 3, pp. 411-430, doi: [10.1332/0305573054325756](https://doi.org/10.1332/0305573054325756).
- Çelikay, F. (2023), "Social spending and chronic unemployment: evidence from OECD countries", *Review of Economics and Political Science*, Vol. 8 No. 2, pp. 86-107, doi: [10.1108/REPS-02-2022-0020](https://doi.org/10.1108/REPS-02-2022-0020).
- Detraz, N. and Peksen, D. (2018), "'Women friendly' spending? Welfare spending and women's participation in the economy and politics", *Politics and Gender*, Vol. 14 No. 2, pp. 137-161, doi: [10.1017/S1743923X17000253](https://doi.org/10.1017/S1743923X17000253).
- Dreher, A. (2006a), "Does globalization affect growth? Evidence from a new index of globalization", *Applied Economics*, Vol. 38 No. 10, pp. 1091-1110, doi: [10.1080/00036840500392078](https://doi.org/10.1080/00036840500392078).
- Dreher, A. (2006b), "The influence of globalization on taxes and social policy - an empirical analysis for OECD countries", *European Journal of Political Economy*, Vol. 22 No. 1, pp. 179-201, doi: [10.1016/j.ejpolco.2005.04.006](https://doi.org/10.1016/j.ejpolco.2005.04.006).
- Dreher, A., Sturm, J.E. and Ursprung, H.W. (2008), "The impact of globalization on the composition of government expenditures: evidence from panel data", *Public Choice*, Vol. 134 Nos 3-4, pp. 263-292, doi: [10.1007/s11127-007-9223-4](https://doi.org/10.1007/s11127-007-9223-4).
- Enns-Jedenastik, L. (2017), "How women's political representation affects spending on family benefits", *Journal of Social Policy*, Vol. 46 No. 3, pp. 563-581, doi: [10.1017/S0047279416000933](https://doi.org/10.1017/S0047279416000933).
- Garrett, G. and Mitchell, D. (2001), "Globalization, Government spending and taxation in the OECD", *European Journal of Political Research*, Vol. 39 No. 2, pp. 145-177, doi: [10.1111/1475-6765.00573](https://doi.org/10.1111/1475-6765.00573).
- Görg, H., Molana, H. and Montagna, C. (2009), "Foreign direct investment, tax competition and social expenditure", *International Review of Economics and Finance*, Vol. 18 No. 1, pp. 31-37, doi: [10.1016/j.iref.2008.02.009](https://doi.org/10.1016/j.iref.2008.02.009).
- Gygli, S., Haelg, F., Potrafke, N. and Sturm, J.E. (2019), "The KOF globalisation index – revisited", *The Review of International Organizations*, Vol. 14 No. 3, pp. 543-574, doi: [10.1007/s11558-019-09344-2](https://doi.org/10.1007/s11558-019-09344-2).
- Haelg, F. (2020), "The KOF globalisation index – a multidimensional approach to globalisation", *Jahrbücher für Nationalökonomie und Statistik*, Vol. 240 No. 5, pp. 691-696, doi: [10.1515/jbnst-2019-0045](https://doi.org/10.1515/jbnst-2019-0045).

- Haelg, F. (2022), "Drivers of financial globalisation: the role of informational frictions", *The World Economy*, Vol. 45 No. 3, pp. 608-636, doi: [10.1111/twec.13181](https://doi.org/10.1111/twec.13181).
- Hayes, A.F. and Cai, L. (2007), "Using heteroskedasticity-consistent standard error estimators in OLS regression: an introduction and software implementation", *Behavior Research Methods*, Vol. 39 No. 4, pp. 709-722, doi: [10.3758/BF03192961](https://doi.org/10.3758/BF03192961).
- Huber, E., Mustillo, T. and Stephens, J.D. (2008), "Politics and social spending in Latin America", *The Journal of Politics*, Vol. 70 No. 02, pp. 420-436, doi: [10.1017/s0022381608080407](https://doi.org/10.1017/s0022381608080407).
- IMF (2020), "Social spending for inclusive growth in the middle east and central Asia", Middle East and Central Asia Department, Paper No, 2020/012.
- Jeong, H. (2010), "Globalization and the Politics of the Welfare State", Phd, Diss, University of Kentucky, available at: https://uknowledge.uky.edu/cgi/viewcontent.cgi?article=1026&context=gradschool_diss
- Karaalp-Orhan, H.S. (2017), "What are the trends in women's labour force participation in Turkey?", *European Journal of Sustainable Development*, Vol. 6 No. 3, pp. 303-312, doi: [10.14207/ejsd.2017.v6n3p303](https://doi.org/10.14207/ejsd.2017.v6n3p303).
- Karaalp-Orhan, H.S. (2018), "The impact of gender-specific human capital on economic growth: an empirical investigation for Turkey", *Zagreb International Review of Economics and Business*, Vol. 21 No. S1, pp. 15-30, doi: [10.2478/zireb-2018-0021](https://doi.org/10.2478/zireb-2018-0021).
- Kaufman, R.R. and Segura-Ubiergo, A. (2001), "Globalisation, domestic politics and social spending in Latin America: a time-series cross-section analysis, 1973-1997", *World Politics*, Vol. 53 No. 4, pp. 553-587, doi: [10.1017/s0043887100019158](https://doi.org/10.1017/s0043887100019158).
- Kaufmann, D., Kraay, A. and Zoido-Lobaton, P. (1999), "Governance matters", Policy Research Working Paper No.2196, World Bank.
- Kaymaz, V. (2018), "OECD ülkelerinde sosyal harcamaların belirleyicileri", *Gazi İktisat ve İşletme Dergisi*, Vol. 4 No. 2, pp. 118-130, doi: [10.30855/gjeb.2018.4.2.005](https://doi.org/10.30855/gjeb.2018.4.2.005).
- Kim, T.K. and Zurlo, K. (2009), "How does economic globalisation affect the welfare state? Focusing on the mediating effect of welfare regimes", *International Journal of Social Welfare*, Vol. 18 No. 2, pp. 130-141, doi: [10.1111/j.1468-2397.2008.00575.x](https://doi.org/10.1111/j.1468-2397.2008.00575.x).
- Kittel, B. and Obinger, H. (2003), "Political parties, institutions, and the dynamics of social expenditure in times of austerity", *Journal of European Public Policy*, Vol. 10 No. 1, pp. 20-45, doi: [10.1080/1350176032000046912](https://doi.org/10.1080/1350176032000046912).
- Knack, S. and Keefer, P. (1995), "Institutions and economic performance: cross-country tests using alternative institutional measures", *Economics and Politics*, Vol. 7 No. 3, pp. 207-227, doi: [10.1111/j.1468-0343.1995.tb00111.x](https://doi.org/10.1111/j.1468-0343.1995.tb00111.x).
- La Porta, R., Silanes, F.L., Schleifer, A. and Vishny, R. (1999), "The quality of government", *Journal of Law, Economics and Organization*, Vol. 15 No. 1, pp. 222-279, doi: [10.1093/jleo/15.1.222](https://doi.org/10.1093/jleo/15.1.222).
- Leibrecht, M., Klien, M. and Onaran, O. (2011), "Globalisation, welfare regimes and social protection expenditures in Western and Eastern European Countries", *Public Choice*, Vol. 148 Nos 3-4, pp. 569-594, doi: [10.1007/s11127-010-9685-7](https://doi.org/10.1007/s11127-010-9685-7).
- Lin, S.S. (1992), "The effect of inflation on government expenditure: new evidence", *Atlantic Economic Journal*, Vol. 20 No. 4, pp. 86, doi: [10.1007/BF02300094](https://doi.org/10.1007/BF02300094).
- Lindert, P.H. (1994), "The rise of social spending", *Explorations in Economic History*, Vol. 31 No. 1, pp. 1-37, doi: [10.1006/exeh.1994.1001](https://doi.org/10.1006/exeh.1994.1001).
- Lindert, P.H. (1996), "What limits social spending?", *Explorations in Economic History*, Vol. 33 No. 1, pp. 1-34, doi: [10.1006/exeh.1996.0001](https://doi.org/10.1006/exeh.1996.0001).
- Lora, E. and Olivera, M. (2007), "Public debt and social expenditure: friends or foes?", *Emerging Markets Review*, Vol. 8 No. 4, pp. 299-310, doi: [10.1016/j.ememar.2006.12.004](https://doi.org/10.1016/j.ememar.2006.12.004).
- Magazzino, C. (2016), "The political determinants of social expenditure in the European countries", *International Journal of Economic Perspectives*, Vol. 10 No. 2, pp. 5-20.

- Magazzino, C., Giolli, L. and Marco, M. (2015), "Wagner's law and peacock and wiseman's displacement effect in European union countries: a panel data study", *International Journal of Economics and Financial Issues*, Vol. 5 No. 3, pp. 812-819.
- Martin, L. and Algarhi, A.S. (2023), "The effect of government spending on economic growth: evidence from the OECD countries", SSRN, available at: <https://ssrn.com/abstract=4337625> or <http://dx.doi.org/10.2139/ssrn.4337625> (accessed 25 January 2023).
- Martín-Mayoral, F. and Nabernegg, M. (2014), *Determinants of Social Spending in Latin America: A Dynamic Panel Data Error-Correction Model Analysis*, 1st ed., Encuentro de Economía Pública, University of Girona, pp. 1-37, available at: <https://dialnet.unirioja.es/servlet/articulo?codigo=5191814>
- Martín-Mayoral, F. and Sastre, J.F. (2017), "Determinants of social spending in Latin America during and after the Washington consensus: a dynamic panel error-correction model analysis", *Latin American Economic Review*, Vol. 26 No. 10, pp. 1-32, doi: [10.1007/s40503-017-0053-6](https://doi.org/10.1007/s40503-017-0053-6).
- Molina-Morales, A., Amate-Fortes, I. and Guarnido-Rueda, A. (2011), "Economic and institutional determinants in fiscal pressure: an application to the European case", *Journal of Economic Issues*, Vol. 45 No. 3, pp. 573-592, doi: [10.2753/JEI0021-3624450303](https://doi.org/10.2753/JEI0021-3624450303).
- Onaran, Ö. and Boesch, V. (2014), "The effect of globalization on the distribution of taxes and social expenditures in Europe: do welfare state regimes matter?", *Environment and Planning a: Economy and Space*, Vol. 46 No. 2, pp. 373-397, doi: [10.1068/a45370](https://doi.org/10.1068/a45370).
- Opler, T.C. (1988), "The effect of inflationary pressure on government expenditure", *Economics Letters*, Vol. 28 No. 4, pp. 331-334, doi: [10.1016/0165-1765\(88\)90008-0](https://doi.org/10.1016/0165-1765(88)90008-0).
- Park, S.S. (2017), "Gendered representation and critical mass: women's legislative representation and social spending in 22 OECD countries", *Sociological Perspectives*, Vol. 60 No. 6, pp. 1097-1114, doi: [10.1177/0731121417710458](https://doi.org/10.1177/0731121417710458).
- Pesaran, M.H. (2004), "General Diagnostic Tests for Cross Section Dependence in Panels", *CESIFO WORKING PAPER NO. 1229*, pp. 255-260, available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=572504
- Pesaran, M.H. and Yamagata, T. (2008), "Testing slope homogeneity in large panels", *Journal of Econometrics*, Vol. 142 No. 1, pp. 50-93, doi: [10.1016/j.jeconom.2007.05.010](https://doi.org/10.1016/j.jeconom.2007.05.010).
- Santos, M. and Simões, M. (2021), "Globalisation, welfare models and social expenditure in OECD countries", *Open Economies Review*, Vol. 32 No. 5, pp. 1063-1088, doi: [10.1007/s11079-021-09646-2](https://doi.org/10.1007/s11079-021-09646-2).
- Schuknecht, L. (1998), "Fiscal policy cycles and public expenditure in developing countries", World Trade Organization (WTO), Geneva, WTO Staff Working Paper No: ERAD-98-06.
- Schuknecht, L. and Zemanek, H. (2018), "Social dominance", CESifo Working Paper Series No, 6894.
- Shabbir, G., Avwar, M. and Shahid, A. (2016), "Corruption, political stability and economic growth", *The Pakistan Development Review*, Vol. 55 No. 4, pp. 689-702, doi: [10.30541/v55i4-iipp.689-702](https://doi.org/10.30541/v55i4-iipp.689-702), available at: <http://www.jstor.org/stable/44986010>
- Snyder, J.M. and Yackovlev, I. (2000), *Political and Economic Determinants of Changes in Government Spending on Social Protection Programs*, Massachusetts Institute of Technology, Cambridge.
- Soroka, S., Johnston, R. and Banting, K. (2006), "Migration and redistribution in a global era", in BardhanBowles, P.S. and Wallerstein (Eds), *Globalization and Egalitarian Redistribution*, Princeton University Press and Russell Sage Foundation, Princeton, NJ, pp. 261-268.
- Tomita, Y. and Kimura, A. (2021), "Financial resource of public social security expenditure, the rule of law, and economic inequality: international comparison of legal origins", *Economic Research-Ekonomska Istraživanja*, Vol. 34 No. 1, pp. 2997-3014, doi: [10.1080/1331677X.2020.1865828](https://doi.org/10.1080/1331677X.2020.1865828).
- Verberi, C. and Yasar, S. (2021), "The effects of social spending on income inequality in 30 OECD countries", *Istanbul Journal of Economics*, Vol. 71 No. 1, pp. 39-57, doi: [10.26650/ISTJECON2021-808121](https://doi.org/10.26650/ISTJECON2021-808121).
- Vergne, C. (2009), "Democracy, elections and allocation of public expenditures in developing countries", *European Journal of Political Economy*, Vol. 25 No. 1, pp. 63-77, doi: [10.1016/j.ejpoleco.2008.09.003](https://doi.org/10.1016/j.ejpoleco.2008.09.003).

- Wagner, A.H. (1883), Finanzwissenschaft. Leipzig: C. F. Winter.
- Woittiez, I. and Kapteyn, A. (1998), "Social interactions and habit formation in a model of female labour supply", *Journal of Public Economics*, Vol. 70 No. 2, pp. 185-205, doi: [10.1016/S0047-2727\(98\)00021-8](https://doi.org/10.1016/S0047-2727(98)00021-8).
- World Bank, (2023a), Government effectiveness, available at: https://govdata360.worldbank.org/indicators/h580f9aa5?country=BRA&indicator=388&viz=line_chart&years=1996,2020 (accessed 19 May 2023).
- World Bank, (2023b), Rule of law, available at: https://govdata360.worldbank.org/indicators/h02f9cb8c?country=BRA&indicator=370&viz=line_chart&years=1996,2020 (accessed 19 May 2023).
- Yay, G.G. and Aksoy, T. (2018), "Globalization and the welfare state", *Quality and Quantity: International Journal of Methodology*, Vol. 52 No. 3, pp. 1015-1040, doi: [10.1007/s11135-017-0501-z](https://doi.org/10.1007/s11135-017-0501-z).

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